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Inventor(s)	Faulhaber; Kurt

Vertebral implants and related methods of use

Abstract

A vertebral insert may include a first linkage, a second linkage, and a third linkage. The first, second, and third linkages may at least partially defining a cavity. The insert may be movable between a collapsed configuration and an expanded configuration, and the movement of the first and second linkages with respect to one another may be configured to reciprocally move the insert between the collapsed and expanded configurations.

Inventors:	Faulhaber; Kurt (Renton, WA)
Applicant:	GLOBUS MEDICAL, INC. (Audubon, PA)
Family ID:	55347284
Assignee:	Globus Medical, Inc. (Audubon, PA)
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Primary Examiner: Sipp; Amy R

Background/Summary

CROSS REFERENCE TO RELATED APPLICATION (1) This application is a continuation of U.S. patent application Ser. No. 16/185,884 which is a divisional application of U.S. application Ser. No. 14/466,640, filed Aug. 22, 2014, which is hereby incorporated by reference in its entirety for all purposes.

FIELD OF THE INVENTION

(1) Various examples of the present disclosure relate generally to vertebral inserts, such as, e.g., fusion devices and related systems and methods. More specifically, the present disclosure relates to vertebral inserts, devices, systems, and methods for repairing and/or replacing intervertebral discs of a patient.

BACKGROUND

(2) A common procedure for handling pain associated with intervertebral discs that have become degenerated due to various factors such as trauma or aging is the use of intervertebral spacers to fuse, e.g., one or more adjacent vertebral bodies. Generally, to fuse the adjacent vertebral bodies, the native intervertebral disc is first partially or fully removed. An intervertebral spacer is then typically inserted between neighboring vertebrae to maintain normal disc spacing and restore spinal stability, thereby facilitating an intervertebral fusion.

(3) There are a number of known conventional intervertebral spacers and methodologies in the art for accomplishing the vertebral fusion. These include screw and rod arrangements, solid bone implants, and intervertebral spacers which include a cage or other implant mechanism which, typically, may be packed with bone and/or bone-growth inducing substances. These devices are implanted between adjacent vertebral bodies in order to fuse the vertebral bodies together, potentially alleviating any associated pain.

(4) However, there are drawbacks associated with the known conventional intervertebral spacers and methodologies. For example, some conventional intervertebral spacers may lack sufficient surface area to optimally fuse adjacent vertebral bodies.

SUMMARY OF THE DISCLOSURE

(5) The present disclosure relates to examples of interbody vertebral inserts and related methods of use.

(6) In one aspect, the present disclosure is directed to a vertebral insert. The vertebral insert may include a first linkage, a second linkage, and a third linkage. The first, second, and third linkages may at least partially defining a cavity. The insert may be movable between a collapsed configuration and an expanded configuration, and the movement of the first and second linkages with respect to one another may be configured to reciprocally move the insert between the collapsed and expanded configurations.

(7) In another aspect, the present disclosure is directed to a method of installing an expandable vertebral insert. The method may include inserting the vertebral insert between two vertebral bodies while the vertebral insert is in a collapsed configuration. The vertebral insert may include a first linkage, a second linkage, and a third linkage. The first, second, and third linkages may at least partially define a cavity. The insert may be movable between a collapsed configuration and an expanded configuration. The movement of the first and second linkages with respect to one another may be configured to reciprocally move the insert between the collapsed and expanded configurations. The method also may include expanding the vertebral insert to an expanded configuration such that a cavity defined by the vertebral insert is larger in the expanded configuration than in the collapsed configuration.

(8) In yet another aspect, the present disclosure is directed to a vertebral insert. The vertebral insert may include a support member having a superior surface, an inferior surface, a first groove disposed between the superior surface and inferior surface on a first lateral side, a second groove disposed between the superior surface and inferior surface on a second lateral side, and a first keel extending from one of the superior and inferior surfaces. The vertebral insert also may include a

first lateral spacer having a portion configured to extend within the first groove, and a second lateral spacer having a portion configured to extend within the second groove.

(9) In yet another aspect, the present disclosure is directed to a vertebral spacer system. The system may include a vertebral spacer, and a bracket coupled to the vertebral spacer. The system also may include a first plate, and a second plate coupled to the first plate and the bracket. The second plate may be movable relative to the first plate, and the second plate may define a cavity through which the vertebral spacer extends.

(10) In yet another aspect, the present disclosure is directed to a method of treating a patient. The method may include attaching a first plate to a first vertebral body, and attaching a second plate to a second vertebral body. The first plate and the second plate may be coupled to one another. The method also may include moving the first and second plates away from one another to distract the first and second vertebral bodies, and inserting a vertebral insert through a cavity defined by the second plate.

(11) In yet another aspect, the present disclosure is directed to a method of treating a patient. The method may include inserting a vertebral insert into the intervertebral space between two adjacent vertebral bodies, and expanding the vertebral insert to distract the adjacent vertebral bodies. The method also may include inserting a spacer into the intervertebral space, and laterally coupling the spacer to the vertebral insert after expanding the vertebral insert.

Description

BRIEF DESCRIPTION OF THE FIGURES

(1) The accompanying drawings, which are incorporated in and constitute a part of this specification, illustrate various exemplary embodiments and together with the description, serve to explain the principles of the disclosed embodiments.

(2) FIG. 1 is a perspective view of a vertebral insert in accordance with an example of the present disclosure.

(3) FIG. 2 is a perspective view of a support member disposed between adjacent vertebral bodies in accordance with an example of the present disclosure.

(4) FIGS. 3-9 depict an exemplary method of installing a vertebral insert in accordance with an example of the present disclosure.

(5) FIG. 10 is an exploded view of a vertebral insert in accordance with an example of the present disclosure.

(6) FIG. 11 is a top view of the vertebral insert of FIG. 10 with an endplate removed.

(7) FIG. 12 is perspective view of a vertebral insert in accordance with an example of the present disclosure.

(8) FIGS. 13-16 depict an exemplary method of installing a vertebral insert in accordance with an example of the present disclosure.

(9) FIGS. 17-20 depict a vertebral insert in accordance with an example of the present disclosure.

(10) FIG. 21 is a perspective view of a spacer assembly in accordance with an example of the present disclosure.

(11) FIGS. 22-24 are perspective views of a plate assembly in accordance with an example of the present disclosure.

(12) FIGS. 25-27 are perspective views of a bracket and spacer in accordance with an example of the present disclosure.

(13) FIGS. 28-36 depict an exemplary method of installing a spacer assembly in accordance with an example of the present disclosure.

DETAILED DESCRIPTION

(14) Reference will now be made in detail to examples of the present disclosure, examples of

which are illustrated in the accompanying drawings. Wherever possible, the same reference numbers will be used throughout the drawings to refer to the same or like parts.

(15) A vertebral insert **100** shown in FIG. **1** may extend from a first end **102** toward a second end **104**, and include a support member **106**, a first lateral insert **108** and a second lateral insert **110**. Support member **106**, and first and second lateral inserts **108**, **110** may be formed from a rigid, bio-compatible material such as, e.g., titanium or polyetheretherketone (PEEK), among others. Support member **106** and first and second lateral inserts **108**, **110** may be formed of the same or different materials. In one example, support member **106** may be formed from titanium while first and second lateral inserts **108**, **110** are formed from PEEK. Components that contact bone, including support member **106**, and first and second lateral inserts **108**, **110**, may be treated with a titanium and/or hydroxyapatite plasma spray coating to encourage bony on-growth, improving the strength and stability of the connection between the respective component and the underlying bone (e.g., a vertebral body). Any other suitable coating also may be provided on one or more surfaces of support member **106**, and first and second lateral inserts **108**, **110**. Such coatings may include therapeutic and/or antibiotic agents, if desired. Support member **106** and first and second lateral inserts **108**, **110** also may include radiopaque markings to facilitate in vivo visualization and insertion.

(16) Support member **106** may have an upper or superior surface **112** and a lower or inferior surface **113**. In some examples, superior surface **112** and inferior surface **113** may be substantially parallel to one another, or may have another suitable relationship to one another. In one example, superior surface **112** and/or inferior surface **113** may have a substantially flat configuration. In some examples, superior surface **112** and/or inferior surface **113** may have a variable cross-section, such as, e.g., bi-convex, e.g., curved from left to right and front to back, or may have another suitable configuration. This curvature may give superior surface **112** and/or inferior surface **113** a partial dome or spherical shape. The curvature may be complementary to the natural curvature of an adjacent vertebral body and may provide for an anatomical fit between superior surface **112** and/or inferior surface **113** with a vertebral body (shown in FIG. **2**). Superior surface **112** and inferior surface **113** may be substantially similar to one another or may have different features. In one example, superior surface **112** and inferior surface **113** may be mirror images of one another. Superior surface **112** may be coupled to inferior surface **113** by a member **116**. Superior surface **112**, inferior surface **113**, and member **116** may generally form an I-beam type configuration defining a first groove **198** and a second groove **199** (referring to FIG. **2**). That is, a groove **198** may be disposed between superior and inferior surfaces **112** and **113** at a first lateral side of vertebral insert **100**, and a groove **199** may be formed between superior and inferior surfaces **112** and **113** at a second lateral side of vertebral insert **100**.

(17) One or more keels **114** may be located on superior surface **112** and may extend away from superior surface **112** at an angle. In one embodiment, the one or more keels **114** may include a plurality of serrations or other geometric configurations. In the example shown in FIG. **1**, keel **114** may be formed at least partially from a substantially elongate member having one or more notches **117** disposed on an axial end surface of keel **114**. However, in some examples, a plurality of keels that are longitudinally or laterally spaced from one another may extend away from superior surface **112**. Once a discectomy has been completed, removing the damaged natural disc or damaged disc material, a groove or channel **180** (shown in FIG. **3**) may be cut into the vertebral body to receive keel **114**. Keel **114** may be integrally formed with superior surface **112** or may alternatively be attached to superior surface **112** by a suitable securing mechanism, e.g., an adhesive. Keel **114** may have one or more holes, e.g., through holes, or openings (not shown) perpendicular to the longitudinal axis of the keel. These holes or openings may provide an aperture for bony in-growth, which may strengthen the connection or interface between the endplate and the vertebral body. In some examples, a plurality of keels **114** may be arranged in a plurality of rows and/or columns, if desired, which can be inserted through a similarly arranged sequence of channels or grooves formed in a

vertebral body. Keels **114** may include suitable coatings or other geometrical features (e.g., barbs or other protrusions) to promote bony in-growth/on-growth. In examples having a plurality of keels **114**, each keel **114** may have varying lengths, widths, and/or other dimensions. One or more keels **115** (which may include one or more of the aforementioned features of keels **114**) may be located on inferior surface **113** and may extend away from inferior surface **113** at an angle.

(18) Keel **114** may include a ramp **120** at the second, leading end **104** of vertebral insert **100**. Superior surface **112** may include two inclined edges directed laterally inward toward a first longitudinal axis A at second, leading end **104**. Longitudinal axis A may extend longitudinally through the longitudinal midsection of superior surface **112**. A first inclined edge **130** of superior surface **112** may incline from a first lateral side of vertebral insert **100** toward longitudinal axis A, while a second inclined edge **132** of superior surface **112** may incline from a second lateral side of vertebral insert **100** toward longitudinal axis A. Ramp **120** and inclined edges **130** and **132** may generally converge at a point **134** disposed along longitudinal axis A. Keel **115** may include a ramp **136** (shown only in FIG. 3) that may include one or more features of ramp **120** of keel **114**. Inferior surface **113** may include a first inclined edge (not shown) and a second inclined edge **138** that may include one or more of the features of first and second inclined edges **130** and **132** of superior surface **112**. The first inclined edge and second inclined edge **138** of keel **115** may converge toward a second longitudinal axis B. Longitudinal axis B may extend longitudinally through the longitudinal midsection of inferior surface **113**. The first inclined edge and second inclined edge **138** of inferior surface **113** and ramp **136** of keel **115** may generally converge at a point **139** that is disposed along longitudinal axis B.

(19) The bladed or wedged formation of second, leading end **104** of vertebral insert **100** may form the leading insertion end of vertebral insert **100** (e.g., the end that enters the intervertebral space first). The bladed or wedged formations may possess a mechanical advantage over other configurations. More specifically, the bladed or wedged formations may reduce the amount of force required to insert vertebral insert **100** into a patient. Further, in some examples, the portion of support member **106** at first end **102** may be the trailing end of support member **106** (e.g., the end that enters the patient second or last). The trailing end of support member **106** may be inserted between vertebral bodies by any suitable mechanism including, e.g., hammering, sliding, pushing, or other mechanisms.

(20) First lateral spacer **108** may include an elongate member **163** that extends from a first, trailing end **160** toward a second, leading end **162**. In some examples, elongate member **163** may be generally cuboid, although other suitable configurations, e.g., cylindrical or the like, are also contemplated. Elongate member **163** may cooperate with groove **198** or **199** to couple support member **106** with first lateral spacer **108**. That is, elongate member **163** and groove **198** or **199** may form, e.g., a tongue and groove joint, although other suitable joints and mating relationships are also contemplated. A generally ellipsoid (or otherwise elongated) ring **165** may be partially defined by flange portions **170** and **172**, and may extend from elongate member **163**. The ellipsoid ring **165** may at least partially define the outer perimeter of a cavity **167**. Cavity **167** may have a generally ellipsoidal (or otherwise elongated) cylindrical shape, although other suitable configurations are also contemplated. In some examples, the outer curvature of ellipsoid ring **165** may be determined by the posterior curvature of the vertebral bodies between which vertebral insert **100** is implanted.

(21) Cavity **167** may be filled with bone graft or similar bone-growth inducing material to further promote and facilitate the intervertebral fusion. Cavity **167**, in one example, may be packed with bone graft or similar bone-growth inducing material to promote the growth of bone through and around the vertebral insert. Such bone graft may be packed into cavity **167**, subsequent to, or during implantation of the vertebral insert **100**.

(22) Second lateral spacer **110** may include an elongate member **143** that extends from a first, trailing end **140** toward a second, leading end **142** (referring to FIG. 4). In some examples, elongate member **143** may be generally cuboid, although other suitable configurations, e.g., cylindrical, are

also contemplated. Elongate member **143** may cooperate with groove **198** or **199** to couple support member **106** with second lateral spacer **110**. That is, elongate member **143** and groove **198** or **199** may form, e.g., a tongue and groove joint, although other suitable joints and mating relationships are also contemplated. Flange portions **144**, **146**, and **148** may extend away from a lateral end surface of elongate member **143** in the superior and inferior directions forming a generally t-shaped cross section with elongate member **143**. In some examples, flange portions **144**, **146**, and **148** may mate and/or cooperate with lateral end surfaces of superior surface **112** and inferior surface **113**, respectively. A groove **150** may be disposed longitudinally between flange portions **144** and **146**, while a groove **152** may be disposed between flange portions **146** and **148**. It should be noted that while two grooves and three flanges are shown in some examples, any other suitable number of grooves and flanges alternatively may be utilized to improve graft area, and/or ease of machining, among other factors. An arc portion **154** may extend laterally outward between first end **140** and second end **142**. Arc portion **154**, flanges **144**, **146**, and **148**, and grooves **150** and **152** may define a cavity **156**. In some examples, flanges **144**, **146**, **148**, and arc portion **154** may define a generally D-shaped ring. It should be noted that, in some examples, the curvature of arc portion **154** may approximate the anterior curvature of adjacent vertebral bodies between which vertebral insert **100** is inserted. Cavity **156** may be filled with bone graft or other bone-growth inducing material in a substantially similar manner as described with respect to cavity **167**.

(23) In some examples, vertebral insert **100** may be delivered by a minimally invasive (MIS) lateral lumbar interbody fusion (LLIF) approach. Vertebral insert **100** may be configured to provide increased surface area over conventional vertebral inserts. Vertebral insert **100** may be self-distracting with the insertion of keels **114**, **115**, and may decrease the chance of subsidence. In some examples, vertebral insert **100** may not need additional fixation such as, e.g., plates or other locking systems because keels **114** and **115** may prevent vertebral insert **100** from backing out of the intervertebral space. That is, vertebral insert **100** may be a standalone implant. Vertebral insert **100** may provide improved spacer fit by allowing for anterior and posterior spacer selection. In some examples, vertebral insert **100** may provide increased total surface area contact with vertebral bodies while also engaging harder bone around the apophyseal ring. In still further examples, vertebral insert **100** may provide for increased amount of bone graft or bone-growth inducing substances to be utilized over conventional devices.

(24) FIGS. 3-9 illustrate an exemplary installation sequence for vertebral insert **100**. Initially, a groove **180** may be formed through the superior surface of a vertebral body **2**. A groove **182** (shown only in FIG. 9) also may be formed in the inferior surface of vertebral body **4**. Keels **115** and **114** of support **106** then may be aligned with grooves **180** and **182**, respectively. Support **106** then may be inserted between vertebral bodies **2** and **4** by a suitable mechanism to place keels **115** and **114** into the grooves formed in the vertebral bodies, and to place superior and inferior surfaces **112** and **113** into the intervertebral space. Once support **106** is secured between vertebral bodies **2** and **4**, one or more lateral inserts (e.g., first and second lateral inserts **108**, **110**) may be aligned with one or more grooves **198**, **199** of support **106** and inserted between vertebral bodies **2** and **4**. It should be noted that any suitable lateral insert may be selected to mate with support **106**. Thus, the use of support **106** may permit operator flexibility and discretion when installing vertebral insert **100**. For example, one or both of first and second lateral spacers **108**, **110** may include modifications to, e.g., outer curvature, height, lordosis correction, or other features depending on the specific geometry of vertebral bodies **2** and **4**. In some examples, lateral spacers **108** and **110** may provide for a continuous amount of lordotic correction, or alternatively, may each be configured to provide a different amount of lordotic correction to various portions of the vertebral bodies. In some examples, vertebral insert **100** may provide a higher amount of height and lordotic correction by having separate anterior and posterior spacers. In some examples, vertebral insert **100** may allow for evenly distributed weight loads. Lateral spacers **108** and **110** may be secured between vertebral bodies **2** and **4** by any suitable mechanism. In one example, rotatable locking

flanges or screws (not shown) may be disposed at the trailing end of support **106**. After lateral spacers **108** and **110** are inserted between the vertebral bodies **2** and **4**, the locking flanges or screws may be rotated into a locking position to prevent spacers **108** and **110** from backing out of the intervertebral space.

(25) FIGS. **10** and **11** show different views of a vertebral insert **200**, in accordance with a further aspect of the present disclosure. The vertebral insert **200** may include a body portion **202**, a first endplate **204**, a second endplate **206**, a translation member **208**, and an actuation member **210**. Translation member **208** and actuation member **210** may cooperate to axially (e.g., vertically) move endplates **204** and **206** away from one another, thereby increasing the axial height of vertebral insert **200**. The first endplate **204** can include a pair of openings **212** and **214** through which bone graft material can be received or deposited. Likewise, the second endplate **206** can have one or more similar openings. In addition to these features, the vertebral insert **200** may include an interference nut **216** that is operably attached to a rear section (e.g., a trailing end) of the body portion **202**, as well as a pair of stabilization members **218**, **220**. In some examples, endplates **204** and **206** may be formed of PEEK or another suitable material.

(26) FIG. **10** illustrates an exploded view of a vertebral insert **200**, while FIG. **11** shows a top view of the same device with a first endplate **204** removed. As shown in both views, a translation member **208** may include at least three expansion portions **222**, **224**, and **226**, which may be connected via bridge portions **227**. The expansion portions **222**, **224**, and **226** each may have angled surfaces that are configured to engage grooved portions of the first and second endplates **204** and **206**. In some examples, the angled surfaces may be of similar angles, while in other examples, the angled surfaces may be of different angles. In at least some examples, providing at least three expansion portions **222**, **224** and **226** may allow for an even expansion along a majority of the length of the body portion **202** of the vertebral insert **200**.

(27) The translation member **208** may be received in the central opening of the body portion **202**. The body portion **202** may include a first end **228** and a second end **230**. In some examples, the first end **228** may include one or more apertures **232**, **234** as shown in FIGS. **10** and **11**. These apertures **232**, **234** may receive one or more stabilization members **218**, **220**.

(28) In some examples, the stabilization members **218**, **220** each may include a first substantially smooth portion **236**, **238**, respectively, and a second threaded portion **240**, **242**, respectively. The stabilization members **218**, **220** may be inserted through the apertures **232**, **234** of the body portion **202**, with the threaded portions **240**, **242** serving as the leading end that enters the apertures. After passing through the apertures **232**, **234** of the body portion **202**, the stabilization members **218**, **220** may come into contact with a side of the translation member **208**. In some examples, the threaded portions **240**, **242** of the stabilization members **218**, **220** may be threaded into mateable threaded surfaces of the translation member **208**. In some examples, by using a pair of stabilization members **218**, **220** as shown in FIGS. **10** and **11** on a first end of the body portion **202**, rocking of the body portion **202** during expansion and contraction of the device **10** may be prevented.

(29) While the illustrated example in FIGS. **10** and **11** show a pair of stabilization members **218**, **220**, in other examples, a single stabilization member or more than two stabilization members can be used to assist in preventing rocking of the body portion **202**. In addition, while the stabilization members **218**, **220** are illustrated as having a substantially cylindrical surface section, in other examples, the stabilization members **218**, **220** can include other shapes and geometries. For example, in other examples, the stabilization members **218**, **220** can have a surface that includes at least one edge or corner.

(30) As shown in FIGS. **10** and **11**, the body portion **202** also may include interference nut **216** that is positioned within a rear section of the body portion **202**. In some examples, the interference nut **216** may be separate and removable from the body portion **202**, while in other examples, the interference nut **216** may not be removable from the body portion **202**. In some examples, the interference nut **216** may include a square nut that is operably connected to a rear section of the

body portion **202**. The interference nut **216** can be mateably connected to a rear of the body portion **202**, for example, via a dove-tail type cut that encapsulates the interference nut **216**. The interference nut **216** can be advantageously formed of a biocompatible material. In some examples, the interference nut **216** may be formed of PEEK.

(31) The interference nut **216** can include a hole (not shown) that is capable of receiving the actuation member **210** therethrough. The actuation member **210**, which can comprise a threaded set screw, may pass through the interference nut **216** and into contact with the translation member **208**, as best shown in FIG. **11**. The interference nut **216** may serve to add drag to the actuation member **210** as it passes therethrough, thereby establishing an interference fit. By providing an interference fit, the risk of the actuation member **210** being loosened prior to or during use may be minimized.

(32) A vertebral insert **300** is shown in FIG. **12** that may be substantially similar to vertebral insert **200** except that a body portion **302** may be utilized instead of body portion **202**. Vertebral insert **300** may include first and second endplates **304** and **306** that may include one or more features of first and second endplates **204** and **206** of vertebral insert **200**. Body portion **302** may be substantially similar to body portion **202** except that an elongate member **308** may extend from a longitudinal side surface **310** of body portion **302**. Elongate member **308** may generally have a T-shaped cross section that extends along a portion of longitudinal side surface **310**, although other suitable configurations are also contemplated. A stabilization aperture **312** may be disposed at a middle portion of elongate member **308**. Stabilization aperture **312** may be configured to receive a threaded portion of a stabilization member (not shown) to couple body portion **302** to a translation member (not shown, but similar to translation member **208** described with reference to FIG. **11**). Thus, stabilization aperture **312**, in at least some examples, may help to provide rotational stability to the translation member relative to the body portion **302** prior to or during use of the vertebral insert **300**.

(33) Once vertebral insert **300** is installed between vertebral bodies as shown in FIGS. **13** and **14**, the superior-inferior height (e.g., the distance between endplates **304** and **306**) may be increased. When vertebral insert **300** is moved to the expanded height, a lateral spacer **340** also may be inserted between vertebral bodies **2** and **4**. In other examples, lateral spacer **340** may be inserted into the intervertebral space and coupled to vertebral insert **300** before the height of vertebral insert **300** is increased. Lateral spacer **340** may be configured to mate with vertebral insert **300**. Lateral spacer **340** may extend from a first, trailing end **342** toward a second, leading end **344**. Lateral spacer **340** may include a groove or channel **346** configured to mate with elongate member **308** of body portion **302**. Channel **346** may be at least partially defined by one or more rails **348**. The ends of channel **346** and rails **348** may terminate at a gap **369** which extends in the superior-inferior directions. Gap **369** may facilitate machining of lateral spacer **340** and may increase the amount of graft that can be applied to vertebral insert **300**. In some examples, any additional number of gaps **369** may be included along the length of channel **346** and rails **348**. That is, channel **346** and rails **348** may include a series of interrupted channels and rails that through which elongate member **308** of body portion **302** may pass through. In some examples, elongate member **308**, channel **346**, and rails **348** may form a tongue and groove joint, although other suitable connections are also contemplated. Gap **369** may be configured to separate channel **346** and rails **348** from a stop surface **364** that is configured to abut elongate member **308** of vertebral insert **300**. In some examples, lateral spacer **340** may define a cavity **350** that may be similar to cavities **150** and **167** described above. Cavity **350** may be packed with bone graft or bone-growth inducing substances in a substantially similar manner as cavities **150** and **167**. In some examples, cavity **350** may be at least partially defined by an arc portion **352** that may approximate the anterior curvature of a vertebral body. However, any or all portions of vertebral insert **300** can be formed in any other suitable configuration depending upon the geometry of the vertebral bodies between which it is to be inserted.

(34) In one example, body portion **302** may be positioned at the posterior end between two

vertebral bodies. However, it is also contemplated that body portion **302** also may be positioned toward the anterior end. Then, lateral spacer **340** may be inserted at the anterior end (if body portion **302** is positioned at the posterior end) or the posterior end (if body portion **302** is positioned at the anterior end). It is also contemplated that body portion **302** may include an additional elongate member **308** at a longitudinal side surface opposite of longitudinal side surface **310** to allow body portion **302** to mate with two different lateral spacers. For example, in some examples, body portion **302** may accommodate an additional lateral spacer, e.g., as shown in FIGS. **1-9**. FIG. **15** shows a top view of vertebral insert **300** and lateral spacer **340** installed on a vertebral body. FIG. **16** is a side view of vertebral insert **300** and lateral spacer **340** installed between vertebral bodies. In some examples, one or more of vertebral insert **300** and lateral spacer **340** may be configured to provide lordosis correction. In some examples, vertebral insert **300** and lateral spacer **340** together may form a continuous angled and ramped surface. In other examples, vertebral insert **300** and lateral spacer **340** each may be ramped at different angles to provide differing degrees of lordosis correction based on the specific geometries of vertebral bodies between which they are inserted.

(35) In some examples, vertebral insert **300** and lateral spacer **340** may be secured between vertebral bodies **2** and **4** via a minimally invasive plate system that is configured to provide stabilization through a lateral approach. The plate system may include brackets, bone screws, blocking screws, and other suitable elements to maintain vertebral insert **300** and lateral spacer **340** between vertebral bodies **2** and **4**.

(36) In some examples, vertebral insert **300** and lateral spacer **340** may be delivered by a minimally invasive (MIS) lateral lumbar interbody fusion (LLIF) approach. Vertebral insert **300** and lateral spacer **340** may be configured to provide increased surface area over conventional vertebral inserts. Vertebral insert **300** may provide improved spacer fit by allowing for anterior and/or posterior spacer selection. In some examples, vertebral insert **300** and lateral spacer **340** may provide increased total surface area contact with vertebral bodies while also engaging harder bone around the apophyseal ring. In still further examples, vertebral insert **300** and lateral spacer **340** may provide for increased amount of bone graft or bone-growth inducing substances to be utilized over conventional devices. In some examples, vertebral insert **300** may provide for more even weight load distribution, reduce cage subsidence, provide a large fusion bed graft area, and/or a higher degree of height and lordotic correction by having separate posterior and/or anterior spacers.

(37) Another example of a vertebral insert **400** is shown in FIGS. **17-20**. Vertebral insert **400** may be reciprocally movable between a first, collapsed configuration (shown in FIG. **17**) and a second, expanded configuration (shown in FIG. **18**). In some examples, vertebral insert **400** may be a collapsible and expandable ring-shaped member that defines a cavity **460**. Cavity **460** may be packed with bone graft in a substantially similar manner as set forth above. Vertebral insert **400** may extend from a first, trailing end **402** toward a second, leading end **404**. Vertebral insert **400** may include a first linkage **406**, a second linkage **408**, a third linkage **410**, and a fourth linkage **412**.

(38) First linkage **406** may include a hollow elongate portion **414** and a flange portion **416**. Hollow elongate portion **414** may be generally cylindrical or have another suitable shape. Flange portion **416** may extend radially outward from hollow elongate portion **414** at a middle longitudinal section or other suitable longitudinal section. An elongate member **418** (e.g., a rod) may extend from flange portion **416** to couple first linkage **406** with second linkage **408**. Elongate member **418** also may prevent the rotation of first linkage **406** relative to second linkage **408**. In an alternative example, hollow elongate portion **414** may be formed in a shape configured to prevent the relative rotation of the first and second linkages **406** and **408** (such as, e.g., square, rectangular, star, or the like). Flange portion **416** also may include features **417** to couple first linkage **406** to third linkage **410**. Features **417** may include a pair of flanges configured to form a joint with an extension of third linkage **410** (e.g., a bridle joint, a mortise and tenon joint, or another suitable joint or connection). In some examples, the pair of flanges may have apertures or pin holes that are aligned

with one another to receive a securing mechanism such as, e.g., a pin that extends through each of the pair of flanges and the extension of third linkage **410**, so as to rotatably couple first linkage **406** with third linkage **410**.

(39) Second linkage **408** may include an end portion **420** having a recess **422** (referring to FIG. **18**) disposed therethrough that is configured to slidably receive hollow elongate portion **414** of first linkage **406**. A longitudinal portion **424** may extend from end portion **420** toward a distraction end **426**. End portion **420** may be disposed at an angle (e.g., 90 degrees) relative to longitudinal portion **424**. A recess **427** (shown in dotted lines in FIG. **17**) may extend through end portion **420** and at least partially through longitudinal portion **424** to receive elongate member **418** therein. Distraction end **426** may be disposed at second, leading end **404** of vertebral insert **400** and may be wedged, bladed, or have another suitable shape to facilitate the distraction of a pair of vertebral bodies upon insertion of vertebral insert **400**. That is to say, distraction end **426** may include a tapered configuration. In some examples, because of the relationship of recess **422** and elongate portion **414**, and of elongate member **418** and recess **427**, first linkage **406** and second linkage **408** may be configured to slidably and longitudinally move with respect to one another.

(40) Third linkage **410** may be a generally elbow shaped member that may be coupled to first linkage **406** (via features **417** described above) at one end, and coupled to fourth linkage **412** at a second end. Third linkage **410** may be formed from two arms **428** that converge at a bend **430** joining the two arms **428**. In some examples, the two arms **428** may form an oblique angle with respect to one another. The second end of third linkage **410** may include mating features **432** that may be substantially similar to mating features **417** of first linkage **406**. The mating features **432** may be configured to receive an extension of fourth linkage **412** to form a joint such as, e.g., a bridle type joint, a mortise and tenon type joint, or another suitable joint or connection.

(41) Fourth linkage **412** also may be a generally elbow shaped member that is coupled to third linkage **410** at one end, and coupled to second linkage **408** at another end. Each end of fourth linkage **412** may include any suitable connector, extension, flange, coupling, or the like to mate with a corresponding feature on third linkage **410** and/or second linkage **408**. Third and fourth linkages **410** and **412** together, when vertebral insert **400** is in the expanded configuration, may form an arc portion that generally approximates the anterior curvature of a vertebral body. In the collapsed configuration of vertebral insert **400**, third and fourth linkages **410** and **412** may be buckled laterally inward. In the expanded configuration of vertebral insert **400**, third and fourth linkages **410** and **412** may bow laterally outward. In some examples, the superior-inferior height of vertebral insert **400** may remain unchanged in both the collapsed and expanded configuration of vertebral insert **400**.

(42) The various linkages described with reference to FIG. **17** may be formed of any suitable biocompatible material, such as, e.g., titanium, PEEK, or the like. In some examples, the linkages may be treated with a titanium and/or hydroxyapatite plasma spray coating to encourage bony on-growth, improving the strength and stability of the connection between the respective component and the underlying bone (e.g., a vertebral body).

(43) Vertebral insert **400** may be inserted between vertebral bodies while in a collapsed configuration (shown in FIG. **17**) while attached to a positioning tool **490**, to another suitable tool, or unattached to a tool. A holding member **492** may be disposed at a distal end of positioning tool **490** and may extend through hollow elongate portion **414** and end portion **420** to couple to a recess or bore (not shown) disposed within distraction end **426** of second linkage **408**. In one example, holding member **492** may be a threaded screw extending from a proximal portion **484**. Holding member **492** may be fastened to distraction end **426** in some examples. An elongate drive member **496** may be slidable relative to holding member **492**. In some embodiments, elongate drive member **496** may be hollow. Once holding member **492** is fastened to distraction end **426** of second linkage **408**, driving member **496** then may be coupled to a first end of first linkage **406** as shown in FIG. **17** (via, e.g., threads or the like). Drive member **496** may be advanced distally to push first

linkage **406** from first end **402** toward second end **404** of vertebral insert **400**, causing vertebral insert **400** to move from the first, collapsed configuration to the second, expanded configuration (FIG. 18). Once vertebral insert **400** is in the expanded configuration, a suitable locking mechanism such as, e.g., a tang, blocking screw, latch, or other suitable locking mechanism may stabilize vertebral insert **400** into the expanded configuration. Once locked into the expanded configuration, holding member **492** and drive member **496** may be disconnected and withdrawn from the patient. In some examples, a distal portion of holding member **492** may be left within vertebral insert **400** and only proximal portion **494** and drive member **496** may be withdrawn. Thus, in some examples, holding member **492** may be configured to lock vertebral insert into the expanded configuration, via, e.g., a rotatable locking flange (not shown) disposed at a proximal end of holding member **492**. Subsequently, bone graft may be packed into cavity **460** via, e.g., a lumen of elongate drive member **496**.

(44) A spacer assembly **500** is shown in FIGS. 21-36. Spacer assembly **500** may include a plate assembly **600**, one or more fastening members **502**, a bracket **800**, and a spacer **802**. As best seen in FIGS. 22 and 23, plate assembly **600** may extend from a first end **602** toward a second **604**. In some embodiments, plate assembly **600** may include a plurality of plates movable relative to each other, as described below. For example, a first plate **606** may extend from first end **602** toward second end **604**, while a second plate **608** may extend from second end **604** toward first end **602**. In embodiments having a single plate (not shown), however, a cavity extending through plate assembly **600** may be selectively adjustable.

(45) With specific reference to FIGS. 22-24, first plate **606** may include a substantially rectangular portion **610** and a substantially semi-circular portion **612** extending from rectangular portion **610**. As shown in FIG. 22, first plate **606** may further include a throughbore **614**. Throughbore **614** may be disposed completely in semi-circular portion **612**, completely in rectangular portion **610**, or partially in semi-circular portion **612** and partially in rectangular portion **610**, as shown. On a first side **616** of first plate **606**, throughbore **614** may include an opening **618**. Opening **618** may be surrounded by a raised lip **620** extending around the periphery of opening **618**. Raised lip **620** may include any suitable configuration. In one embodiment, raised lip **620** may include a substantially square cross-sectional configuration. Those of ordinary skill in the art, however, will recognize that raised lip **620** may include any suitable cross-sectional configuration. Though the depicted embodiment shows raised lip **620** as continuously extending around opening **618**, raised lip **620** may include one or more breaks, such that it does not extend continuously around opening **618**. In some embodiments, opening **618** may further include a radially inward lip **622** extending away from a wall defining opening **618**.

(46) With continued reference to FIG. 22, first side **616** may further include a stepped portion **624**. Stepped portion **624** may be spaced from a surface **626** by a vertical wall **628** (shown in FIG. 24). Stepped portion **624** may include any suitable configuration. In one embodiment, stepped portion **624** may include a substantially constant cross-sectional dimension. However, in some embodiments, stepped portion **624** may include a tapered cross-sectional configuration. Stepped portion **624** may have a smaller cross-sectional dimension (e.g., thickness) relative to a remainder of first plate **606**. Stepped portion **624** may include a surface on first side **616** having one or more geometric configurations **630** to increase friction. For example, geometric configurations **630** may include one or more teeth, ridges, divots, cavities, or other similar structures. Further, stepped portion **624** may define an elongated channel **632** extending through stepped portion **624**. Although a single elongate channel **632** is depicted, stepped portion **624** may include a plurality of channels or cavities. Elongate channel **632** may include any suitable configuration. As will be described in further detail below, elongate channel **632** may be configured to receive a portion of a locking member **634**.

(47) With reference now to FIG. 23, first plate **606** may include a second side **636** disposed opposite to the first side **616**. Second side **636** may be substantially planar. In some embodiments,

an opening **638** of throughbore **614** may be surrounded by a raised atraumatic lip **640**. Lip **640** may include any suitable configuration, including, but not limited to, a curved outer periphery. A portion of lip **640** may include a cutout **642** corresponding to an end of elongate channel **632**. As can also be seen in FIG. 23, elongate channel **632** may extend through first plate **606** to an opening **644** on second side **636**. Opening **644** may include counterbore features extending around a periphery of opening **644** to receive an end of a locking member **634**.

(48) With continued reference to FIGS. 22-24, second plate **608** may include a substantially rectangular portion **650** and a substantially semi-circular portion **652** extending from rectangular portion **650**. Second plate **608** may further include a throughbore **654**. Throughbore **654** may be disposed completely in semi-circular portion **652**, completely in rectangular portion **650**, or partially in semi-circular portion **2215** and partially in rectangular portion **650**, as shown. On a first side **656** of second plate **608**, throughbore **654** may include an opening **658**. Opening **658** may be surrounded by a raised lip **660** extending around the periphery of opening **658**. Raised lip **660** may include any of the features described above with respect to raised lip **620**. In some embodiments, opening **658** may further include a radially inward lip **662** extending away from a wall defining opening **658**.

(49) In the embodiment shown in FIGS. 22-24, rectangular portion **650** may define a cavity **664** (e.g., a rectangular cavity configured to receive a vertebral spacer **802** shown in FIGS. 25-27). Locking member **634** may extend through a recess disposed in an end wall **666** that at least partially defines cavity **664**. The end wall **636** may include a protrusion **668** that extends into cavity **664**. The protrusion **668** may accommodate various features (e.g., locking features) associated with locking member **634**. In some examples, locking member **634** may include a head portion **670** that abuts the first side **656** of second plate **608**. Cavity **664** also may be at least partially defined by a side wall **672** and side wall **674**. In some examples, side walls **672** and **674** may be substantially parallel to one another and orthogonal to end wall **636**. A locking member **676** may extend through a recess disposed in side wall **672** while a locking member **678** may extend through a recess disposed in side wall **674**. Locking members **676** and **678** may include one or more locking features described with reference to locking member **634**. Side wall **672** may include a protrusion **680** and side wall **674** may include a protrusion **682**. Protrusions **680** and **682** may accommodate locking features of locking members **676** and **678**, respectively, and may be configured to extend away from cavity **664** in opposite directions.

(50) With reference now to FIG. 23, second plate **608** may include a second side **686** disposed opposite to the first side **656**. Second side **686** may be substantially planar. In some embodiments, an opening **688** of throughbore **654** may be surrounded by a raised atraumatic lip **690**. Lip **690** may include may include one or more features described above with reference to lip **640**. Second side **686** may further include a stepped portion **694** disposed on end wall **674**. Stepped portion **694** may be spaced from a surface **696** by a vertical wall **698** (shown in FIG. 23). Stepped portion **694** may include any suitable configuration including those described with reference to stepped portion **624** of first plate **606**, and may include geometric features **700** that may be substantially similar to geometric features **630**. Second side **686** also may include a stepped portion **704** that is disposed along end wall **672**. Stepped portion **704** may be spaced from surface **696** by a vertical wall **708** and may include geometric features **710** that are substantially similar to geometric features **630** and **700**. The stepped portions of second plate **608** (e.g., stepped portions **694**, **704** and a stepped portion of end wall **666**) may have a smaller cross-sectional dimension (e.g., thickness) than a remainder of second plate **608**. Side walls **672** and **674** also may include counterbore features **712** and **714** that may be configured to receive portions of locking members **676** and **678**, respectively.

(51) As described above, first and second plates **606** and **608** may be movable relative to one another when coupled via locking member **634**. In some examples, stepped portion **624** of first plate **606** may be slidably coupled to stepped portions **694** and **704** of second plate **608**. When coupled to one another, first and second plates **606** and **608** may form an adjustable joint or other

suitable connection. In one example, first and second plates **606** and **608** may form a half lap splice joint, although other suitable connections are also contemplated.

(52) The relative movement of first plate **606** and **608** with respect to one another may define an effective opening of cavity **664**. That is, cavity **664** may be movable between a first, fully-open configuration (shown in FIG. 24), and a second, partially-open configuration (shown in FIG. 22). In the second, partially-open configuration, stepped portion **624** of first plate **606** may extend over and cover cavity **664**, reducing the effective opening size of cavity **664**. That is, stepped portion **624** may extend over cavity **664** to reduce the size of a vertebral spacer that can be received by cavity **664**. It is also contemplated that cavity **664** may include any number of intermediate configurations between the first, fully-open configuration and the second, partially-open configuration.

(53) Locking member **634** may be reciprocally movable between a plurality of configurations and may be configured to couple first and second plates **606** and **608** to one another. In a first configuration, locking member **634** may couple first plate **606** to second plate **608** and may permit the relative movement of first and second plates **606** and **608** with respect to one another. In a second configuration, locking member **634** may fix the orientation of first plate **606** with respect to second plate **608** by any suitable mechanism such as, e.g., tightening a threaded engagement, moving a locking flange or other mechanism into a locking position, or the like.

(54) FIGS. 25-27 depict exploded and assembled views of a bracket **800** and a spacer **802**. Bracket **800** may be a bracket configured to couple to both spacer **802** and plate assembly **600** described above. Bracket **800** may include a generally rectangular elongate surface **804** having a first elongate cavity **806** and a second elongate cavity **810** extending through elongate surface **804**. The lengths of elongate cavities **806** and **810** may be generally parallel to one another, and each may be generally perpendicular to the length of elongate surface **804**, although other suitable configurations are also contemplated. Bracket **800** also may include a protrusion **812** (shown only in FIG. 26) that extends away from elongate surface **804**. A bore **814** may extend through elongate surface **804** and protrusion **812** to receive a fastener (e.g., a threaded screw or the like). In some examples, bracket **800** may be curved at its lateral edges.

(55) Spacer **802** may be any suitable intervertebral spacer. In the example shown in FIGS. 25-27, spacer **802** may be a generally rectangular spacer defining a cavity **816**. Cavity **816** may be packed with bone graft or bone-growth inducing materials. A lateral end surface of spacer **802** may define a recess **818** and a bore **820** configured to receive protrusion **812** and fastener **822** respectively. Recess **818** and bore **820** of spacer **802** may be complimentary to protrusion **812** and fastener **822** so as to limit movement of bracket **800** and spacer **802** with respect to one another. Spacer **802** may be any suitable intervertebral spacer, and may include one or more of inferior surfaces, superior surfaces, biconvex surfaces, among others. In some examples, the surfaces of spacer **802** or any other bone contacting surface described in the present disclosure may include one or more of teeth, ridges, friction increasing elements, keels, or gripping or purchasing projections.

(56) FIGS. 28-36 depict an exemplary method of installing spacer assembly **500** within a patient. Initially, plate assembly **600** may be brought in proximity with two vertebral bodies **2** and **4** (FIG. 28). In one example, during initial installation, first and second plates **606** and **608** may be in the second, partially-open configuration where cavity **628** has its smallest effective opening. As shown in FIG. 29, bore **654** may be aligned with vertebral body **4** while, bore **614** may be aligned with vertebral body **2**. First and second plates **606** and **608** may be moved toward the fully-opened configuration depending upon the desired distance between vertebral bodies **2** and **4** to optimize fit. It is contemplated that in some examples, plate assembly **600** may be in an intermediate configuration between the first and second configurations at this stage such that cavity **664** is at an intermediate opening size. A fastening member **502** may then be used to secure one of first or second plates **606** and **608** to a respective vertebral body (FIG. 30), and another fastening member **502** may be used to secure another fastener **502** to secure the other of first and second plates **606**

and **608** to a respective vertebral body (FIG. **31**).

(57) Once first and second plates **606** and **608** are secured to vertebral bodies **2** and **4**, an operator may distract vertebral bodies **2** and **4** from one another via first and second plates **606** and **608**. That is, an operator may move first and second plates **606** and **608** away from one another to move vertebral bodies **2** and **4** away from one another. Once vertebral bodies **2** and **4** have been moved away from one another (e.g., distracted) by a sufficient distance, an operator may lock the distracted position by moving locking member **634** from an unlocked configuration to a locked configuration, thereby locking the positions of first and second plates **606** and **608** and also vertebral bodies **2** and **4**. It is further contemplated that the positions of first and second plates **606** and **608** may be fixed with respect to one another by any other suitable mechanism.

(58) Once vertebral bodies **2** and **4** are distracted and locked into position, spacer **802** may be inserted through cavity **664** (FIGS. **34** and **35**). As spacer **802** is moved between the vertebral bodies, bracket **800** may abut second plate **608** of plate assembly **600**. Further, locking elements **676** and **678** may be positioned to extend through elongate cavities **806** and **810**. The locking elements **676** and **678** may then be moved from an insertion configuration to a locking configuration, thereby securing spacer **802** between vertebral bodies **2** and **4**.

(59) Spacer assembly **500** may provide one or more of the following advantages, including: an increased tactile feel while distracting, an increased range of distraction height, increased bone graft size and placement, improved radiolucent viewing, decreased stress on endplates and reduced chance of subsidence, and the potential elimination of the need for posterior fixation.

(60) Any aspect set forth in any example may be used with any other example set forth herein. Every device and apparatus set forth herein may be used in a suitable medical procedure, such as, e.g., a vertebral disc replacement procedure, and may be advanced through any suitable body lumen and body cavity.

(61) It will be apparent to those skilled in the art that various modifications and variations can be made in the disclosed systems and processes without departing from the scope of the disclosure. Other examples of the disclosure will be apparent to those skilled in the art from consideration of the specification and practice of the embodiments disclosed herein. It is intended that the specification and examples be considered as exemplary only.

Claims

1. A vertebral spacer system comprising: a vertebral spacer having a body portion, a first endplate, a second endplate, a translation member having an upper surface, the upper surface having a plurality of angled expansion portions, and an actuation member, wherein the vertebral spacer has a first end configured to be initially inserted into a vertebral space and a second end trailing the first end, wherein the translation member and the actuation member cooperate to move the first and second endplates away from one another and away from the body portion, thereby increasing a height of the vertebral spacer, wherein the vertebral spacer includes an elongate member extending from a longitudinal side surface of the body portion and a stabilization aperture disposed at a middle portion of the elongate member configured to receive a stabilization member to couple the body portion to the translation member, and a lateral spacer having a leading end and a trailing end and a channel configured to slidably mate with the elongate member, wherein the channel terminates with a stop portion to limit travel of the elongate member in the channel.
2. The vertebral spacer system of claim 1, wherein the elongate member is a T-shaped cross section that extends a portion of the longitudinal side surface.
3. The vertebral spacer system of claim 1, wherein the channel of the lateral spacer may include one or more rails.
4. The vertebral spacer system of claim 3, wherein the channel includes a gap which extends in a superior-inferior direction proximate the stop portion.

5. The vertebral spacer system of claim 3, wherein the elongate member, channel and rails of the lateral spacer form a tongue and groove joint.
 6. The vertebral spacer system of claim 1, wherein the lateral spacer includes a cavity for receiving bone graft.
 7. The vertebral spacer system of claim 1, wherein the vertebral spacer and the lateral spacer may form a continuous angled and ramped surface.
 8. The vertebral spacer system of claim 1, wherein: the first endplate has a front end and a rear end, wherein the translation member and the actuation member cooperate to move both the front and rear ends of the first endplate away from the second endplate.
 9. A vertebral spacer system comprising: a vertebral spacer having a body portion, a first endplate, a second endplate, a translation member having an upper surface, the upper surface having a plurality of angled expansion portions, and an actuation member, wherein the vertebral spacer has a first end configured to be initially inserted into a vertebral space and a second end trailing the first end, wherein the translation member and the actuation member cooperate to move the first and second endplates away from one another and away from the body portion, thereby increasing a height of the vertebral spacer, wherein the vertebral spacer includes an elongate member extending from a longitudinal side surface of the body portion and a stabilization aperture disposed at a middle portion of the elongate member configured to receive a stabilization member to couple the body portion to the translation member, and a lateral spacer having a leading end, a trailing end, a first side wall and a second side wall, the first and second side walls connecting the leading end and the trailing end, wherein the first sidewall of the lateral spacer includes a channel configured to slidably mate with the elongate member, wherein the channel terminates with a stop portion to limit travel of the elongate member in the channel, and wherein the second side wall of the lateral spacer is curved forming an arc that is configured to approximate a curvature of a vertebral body.
 10. The vertebral spacer system of claim 9, wherein the elongate member is a T-shaped cross section that extends a portion of the longitudinal side surface.
 11. The vertebral spacer system of claim 9, wherein the channel of the lateral spacer may include one or more rails.
 12. The vertebral spacer system of claim 11, wherein the elongate member, channel and rails of the lateral spacer form a tongue and groove joint.
 13. The vertebral spacer system of claim 9, wherein the lateral spacer includes a cavity for receiving bone graft.
 14. The vertebral spacer system of claim 9, wherein the vertebral spacer and the lateral spacer may form a continuous angled and ramped surface.
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